

CS6101 PA1 Report

Deevyanshu Malu (22B1029)
Deeptanshu Malu (22B1256)
Sai Sriram Sambaraju (22B0940)

September 22, 2025

1 Initial Setup

The relevance score was taken as the BM25 score between the query and the document. The similarity score was taken as the cosine similarity between the BM25 vectors of the documents.

2 Taskwise Metrics

Dataset	K	FL@K	WC@0.1	WC@0.5	WC@0.9
arguana	5	1232.45	0.6061	0.0102	0.0049
	10	1352.94	0.6657	0.0168	0.0092
	20	1507.89	0.7323	0.0272	0.0168
kialo	5	1160.02	0.6115	0.0097	0.0044
	10	1268.35	0.6655	0.0162	0.0086
	20	1410.00	0.7251	0.0268	0.0159
opinionqa	5	1383.14	0.5642	0.0068	0.0040
	10	1516.38	0.6332	0.0121	0.0078
	20	1681.88	0.7070	0.0200	0.0146

Table 1: Results for Task 1 across datasets and different K values

Dataset	K	FL@K	nFL@K	WC@0.1	WC@0.5	WC@0.9
arguana	5	1128.89	0.9142	0.5477	0.0073	0.0044
	10	1249.53	0.9234	0.6111	0.0118	0.0083
	20	1393.96	0.9231	0.6733	0.0201	0.0155
kialo	5	1068.00	0.9152	0.5618	0.0065	0.0041
	10	1178.64	0.9237	0.6244	0.0111	0.0079
	20	1313.40	0.9273	0.6841	0.0196	0.0152
opinionqa	5	1293.25	0.9339	0.5148	0.0051	0.0038
	10	1437.75	0.9463	0.5936	0.0090	0.0072
	20	1602.08	0.9522	0.6652	0.0161	0.0139

Table 2: Results for Task 2 across datasets and different K values

Dataset	K	FL@K	nFL@K	WC@0.1	WC@0.5	WC@0.9
arguana	5	1129.02	0.9143	0.5477	0.0073	0.0044
	10	1249.84	0.9235	0.6112	0.0119	0.0083
	20	1394.66	0.9238	0.6731	0.0203	0.0157
kialo	5	1067.99	0.9153	0.5618	0.0064	0.0041
	10	1178.67	0.9238	0.6243	0.0111	0.0079
	20	1312.66	0.9267	0.6836	0.0195	0.0151
opinionqa	5	1293.31	0.9339	0.5148	0.0051	0.0038
	10	1437.62	0.9464	0.5934	0.0091	0.0073
	20	1602.14	0.9522	0.6653	0.0161	0.0139

Table 3: Results for Task 3 across datasets and different K values

Dataset	K	FL@K	nFL@K	WC@0.1	WC@0.5	WC@0.9
arguana	5	1037.14	0.8454	0.4958	0.0054	0.0043
	10	1177.97	0.8747	0.5861	0.0101	0.0084
	20	1349.88	0.8972	0.6717	0.0195	0.0162
kialo	5	963.14	0.8267	0.4897	0.0051	0.0043
	10	1093.43	0.8595	0.5774	0.0100	0.0084
	20	1254.57	0.8874	0.6587	0.0190	0.0161
opinionqa	5	1153.77	0.8399	0.4099	0.0043	0.0039
	10	1319.17	0.8741	0.5203	0.0084	0.0077
	20	1514.21	0.9028	0.6240	0.0161	0.0148

Table 4: Results for Task 4 across datasets and different K values

Dataset	K	FL@K	nFL@K	WC@0.1	WC@0.5	WC@0.9
arguana	5	1143.30	0.9224	0.5504	0.0073	0.0041
	10	1255.14	0.9244	0.6079	0.0115	0.0077
	20	1389.84	0.9189	0.6650	0.0195	0.0147
kialo	5	1084.80	0.9270	0.5707	0.0067	0.0038
	10	1186.75	0.9295	0.6249	0.0111	0.0074
	20	1310.43	0.9244	0.6786	0.0190	0.0143
opinionqa	5	1316.79	0.9506	0.5301	0.0049	0.0034
	10	1446.15	0.9522	0.5986	0.0084	0.0066
	20	1599.56	0.9501	0.6624	0.0153	0.0131

Table 5: Results for Task 5 across datasets and different K values

3 Task 5 Optimization

Since the objective of this assignment is to get diverse documents which are relevant to the query, we can find this in a very fast way compared to the greedy algo in task 1 by choosing the top K documents based on $\sum_{i \in [N]} r(i) \times s(i, j)$. Where j is our document. The intuition behind this is we have a relevance vector r which represents the similarity of each document to the query and a similarity vector $s(j)$ which represents the similarity of document j to all other documents. By multiplying them we are trying to find documents which are similar to the documents which are relevant to the query. Hence choosing top K documents means choosing documents which are similar to the query. We have optimized this by using a priority queue to find the top K documents instead of iterating over N documents for K times. This reduces the time complexity drastically from $O(N * K)$ to $O(N \log K)$ without significantly compromising with the performance. Below is the table of the results.

Dataset	K	Task 1 Time (s)	Task 5 Time (s)
arguana	5	162.90	153.99
	10	183.21	152.10
	20	214.17	152.77
kialo	5	173.78	156.16
	10	187.30	158.85
	20	223.01	156.02
opinionqa	5	206.48	183.21
	10	216.65	182.43
	20	253.33	181.14

Table 6: Runtime (in seconds) for Task 1 and Task 5 across datasets and different K values

4 Sampled Documents

The docs returned are with $K=5$. The first query from each of the three datasets are taken.

4.1 Task 1

- **Query (Arguana):** "Is ignoring North Korea an effective strategy for resolving conflicts?"
 - Korean conflict The Korean conflict is based on the division between the Democratic People's Republic...
 - Round begins with the following steps. Conflict Resolution: Each player, in Initiative order, resolves their...
 - with large amounts of UN air and naval support, the US and South Korean ground...
 - Foreign relations of North Korea The foreign relations of North Korea – officially the Democratic...
 - 2019 North Korea–United States summit President of the United States Donald Trump has stated that...
- **Query (Kialo):** "What religion should people follow?"
 - and laws should be guided by religion. This last belief, written into the constitutions of...
 - Bahá'í Faith in Mexico The Bahá'í Faith in Mexico begins with visits of Bahá'ís before...
 - other issues of church-state separation in Ireland. He said the Good Friday drink ban is...
 - few come to the notice of humanity, whose belief then shapes and strengthens them until...
 - since Obama is a Christian and has discussed about his faith in the past. Madonna...
- **Query (OpinionQA):** "Do we live in a safer world compared to 50 years ago?"
 - the stores than it was four years ago? Is there more or less unemployment in...
 - ago after the First World War many, many people in my constituency sat in the...
 - site of an ancient lake, and it is a popular place to find fossils of...
 - we don't get it out on the market by the end of this year, it'll...
 - Concern Worldwide Concern Worldwide (often referred to as Concern) is Ireland's largest aid and humanitarian...

4.2 Task 2

- **Query (Arguana):** "Is ignoring North Korea an effective strategy for resolving conflicts?"
 - increased cost of engaging in another overseas military conflict. By instigating the detachment of the...
 - Foreign relations of North Korea The foreign relations of North Korea
 - officially the Democratic...
 - an agreement, exchanging Zimbabwean uranium for DPRK arms. The DPRK is a member of the...
 - Korean conflict The Korean conflict is based on the division between the Democratic People's Republic...
 - with large amounts of UN air and naval support, the US and South Korean ground...
- **Query (Kialo):** "What religion should people follow?"
 - and people who had difficulty living independently in society, such as gangsters. He supported these...
 - other issues of church-state separation in Ireland. He said the Good Friday drink ban is...
 - relationship to it and to any posited entities. In terms of Christian theology and ultimate...
 - only tie to Jesus was celebrating Christmas. Needless to say I had to find my...
 - and laws should be guided by religion. This last belief, written into the constitutions of...
- **Query (OpinionQA):** "Do we live in a safer world compared to 50 years ago?"
 - we do is on display, for better or worse. I think mostly for better, but...
 - sink in. We've been wanting to sit here for 20 years and together we owned...
 - the stores than it was four years ago? Is there more or less unemployment in...
 - no shows, no work on a new album, etc. well. I wanted to let you...
 - ago after the First World War many, many people in my constituency sat in the...

4.3 Task 3

- **Query (Arguana):** "Is ignoring North Korea an effective strategy for resolving conflicts?"
 - increased cost of engaging in another overseas military conflict. By instigating the detachment of the...
 - Foreign relations of North Korea The foreign relations of North Korea
 - officially the Democratic...
 - an agreement, exchanging Zimbabwean uranium for DPRK arms. The DPRK is a member of the...
 - Korean conflict The Korean conflict is based on the division between the Democratic People's Republic...
 - with large amounts of UN air and naval support, the US and South Korean ground...
- **Query (Kialo):** "What religion should people follow?"
 - and people who had difficulty living independently in society, such as gangsters. He supported these...
 - other issues of church-state separation in Ireland. He said the Good Friday drink ban is...
 - relationship to it and to any posited entities. In terms of Christian theology and ultimate...
 - only tie to Jesus was celebrating Christmas. Needless to say I had to find my...
 - and laws should be guided by religion. This last belief, written into the constitutions of...
- **Query (OpinionQA):** "Do we live in a safer world compared to 50 years ago?"
 - we do is on display, for better or worse. I think mostly for better, but...
 - sink in. We've been wanting to sit here for 20 years and together we owned...
 - the stores than it was four years ago? Is there more or less unemployment in...
 - no shows, no work on a new album, etc. well. I wanted to let you...
 - ago after the First World War many, many people in my constituency sat in the...

4.4 Task 4

- **Query (Arguana):** "Is ignoring North Korea an effective strategy for resolving conflicts?"
 - increased cost of engaging in another overseas military conflict. By instigating the detachment of the...
 - such spatially aggregated units, spatial variations of detailed social variables cannot be identified. The problem...
 - Foreign relations of North Korea The foreign relations of North Korea
 - officially the Democratic...
 - cannot co-exist. This can be resolved by a number of techniques, which fall into three...
 - also interpreted this as an expression of a shift in international power structures, as Europe...
- **Query (Kialo):** "What religion should people follow?"
 - and people who had difficulty living independently in society, such as gangsters. He supported these...
 - other issues of church-state separation in Ireland. He said the Good Friday drink ban is...
 - relationship to it and to any posited entities. In terms of Christian theology and ultimate...
 - only tie to Jesus was celebrating Christmas. Needless to say I had to find my...
 - UAE citizens to enjoy free access to them. When asked by "The New York Times"...
- **Query (OpinionQA):** "Do we live in a safer world compared to 50 years ago?"
 - we do is on display, for better or worse. I think mostly for better, but...
 - it's a safer and more enjoyable way to get to Qingdao, and Laixi residents love...
 - sink in. We've been wanting to sit here for 20 years and together we owned...
 - started doing in 2007. This year we've seen that it has caused interest. For us,...
 - girl we would enter an unknown home of unknown people. We would not mention our...

4.5 Task 5

- **Query (Arguana):** "Is ignoring North Korea an effective strategy for resolving conflicts?"
 - Korean conflict The Korean conflict is based on the division between the Democratic People's Republic...
 - Foreign relations of North Korea The foreign relations of North Korea
 - officially the Democratic...
 - an agreement, exchanging Zimbabwean uranium for DPRK arms. The DPRK is a member of the...
 - with large amounts of UN air and naval support, the US and South Korean ground...
 - engagement, the 3rd Battalion, 29th Infantry Regiment was disbanded and merged with other units as...
- **Query (Kialo):** "What religion should people follow?"
 - and laws should be guided by religion. This last belief, written into the constitutions of...
 - and people who had difficulty living independently in society, such as gangsters. He supported these...
 - relationship to it and to any posited entities. In terms of Christian theology and ultimate...
 - moral law, making autonomy incompatible with divine command theory. Robert Adams challenges this criticism, arguing...
 - other issues of church-state separation in Ireland. He said the Good Friday drink ban is...
- **Query (OpinionQA):** "Do we live in a safer world compared to 50 years ago?"
 - the stores than it was four years ago? Is there more or less unemployment in...
 - no shows, no work on a new album, etc. well. I wanted to let you...
 - ago after the First World War many, many people in my constituency sat in the...
 - up with the premise in my living room, where the movie is shot. A couple...
 - we do is on display, for better or worse. I think mostly for better, but...